YUEQIN YIN

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Education

The University of Texas at Austin — Austin, TX
Ph.D. in Artificial Intelligence — *Sept. 2023 - June. 2028(expected)*
- Advisor: Prof. Mingyuan Zhou

CASIA (Institute of Automation, Chinese Academy of Sciences) — Beijing, China
M.S. in Computer Vision and Deep Learning — *Sept. 2020 - June. 2023*

Dalian University of Technology — Dalian, China
B.S. in Software Engineering — *Sept. 2016 - June. 2020*
- Grade: 91.8/100
- 2020 Outstanding Graduate, Liaoning Province

Publications

1. **Yueqin Yin**, Yaxi Li, Xin Liu, Xun Wang, Kaiqiang Song, Simin Ma, Shujian Liu, Sathish Reddy Indurthi, Haoyun Deng, Pengcheng He, Mingyuan Zhou, Song Wang. ***ContextCheck: Sentence-Level Faithfulness Verification with Context-Aware Disambiguation.*** Accepted by **ACL 2026 Findings**.
   - Proposes a context-conditioned faithfulness verifier that resolves pronouns and references by conditioning on preceding discourse, eliminating separate decontextualization steps required by prior pipelines.
   - Achieves SOTA on 3 new context-dependent benchmarks (**10+ point Macro F1 gain**) and matches or surpasses 8B-scale baselines across 14 standard single-sentence datasets (avg. Macro F1: 73.5 vs. 72.8).

2. **Yueqin Yin***, Shentao Yang*, Yujia Xie, Ziyi Yang, Yuting Sun, Hany Awadalla, Weizhu Chen, Mingyuan Zhou. ***Segmenting Text and Learning Their Rewards for Improved RLHF in Language Model.*** Accepted by **Transactions on Machine Learning Research (TMLR)**, 2025. [Paper] [Code] (* denotes equal contribution)
   - Introduces a segment-level reward model that assigns dense feedback to semantically coherent text segments via entropy-based segmentation, bridging sparse sequence-level and overly granular token-level RLHF.
   - Designs location-aware reward normalization and within-segment reward interpolation for stable PPO training, achieving competitive performance on AlpacaEval 2.0, Arena-Hard, and MT-Bench.

3. Zhangchen Xu, Yang Liu, **Yueqin Yin**, Mingyuan Zhou, Radha Poovendran. ***KODCODE: A Diverse, Challenging, and Verifiable Synthetic Dataset for Coding.*** Accepted by **ACL 2025 Findings**; **Best Paper Award at DataWorld @ ICML 2025**. [Paper] [Code] [Website]
   - Introduces a **447K-problem** synthetic coding dataset with self-verified solutions and unit tests spanning 12 diverse sources; each solution achieves 100% branch coverage via automated self-verification (<2.5% error rate).
   - Fine-tuned models (e.g., Qwen2.5-Coder-32B-Instruct) achieve **SOTA on HumanEval(+), MBPP(+), BigCodeBench, and LiveCodeBench**, surpassing DeepSeek-R1-Distill-Llama-70B.

4. Yanbo Xu*, **Yueqin Yin***, Liming Jiang, Qianyi Wu, Chengyao Zheng, Chen Change Loy, Bo Dai, and Wayne Wu. ***TransEditor: Transformer-Based Dual-Space GAN for Highly Controllable Facial Editing.*** In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 7683-7692, 2022. [Paper] [Code] [Website] (* denotes equal contribution)
   - Proposes a Transformer-based dual-space GAN with a **Cross-Space Interaction** mechanism between disentangled P-space and Z-space, enabling independent control of structural (pose) and textural (style) facial attributes.
   - Achieves **SOTA controllability and identity preservation** on CelebA-HQ and FFHQ, outperforming StyleGAN2, StyleMapGAN, and DAT in both qualitative and quantitative evaluations.

5. Shiyue Cao, **Yueqin Yin**, Lianghu Huang, Yu Liu, Xin Zhao, Deli Zhao, Kaiqi Huang. ***Efficient-VQGAN: Towards High-Resolution Image Generation with Efficient Vision Transformers.*** In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 7368-7377, 2023. [Paper]
   - Replaces global attention with **local Swin Transformer blocks** for vector quantization, significantly reducing computational cost while improving reconstruction fidelity for high-resolution image generation.

◇ Introduces a multi-grained attention mechanism and a unified autoencoding-training / autoregressive-inference paradigm, achieving **SOTA fidelity and speed on FFHQ and ImageNet**.

Preprint

1. **Yueqin Yin***, Zhendong Wang*, Yi Gu, Hai Huang, Weizhu Chen, Mingyuan Zhou. ***Relative Preference Optimization: Enhancing LLM Alignment through Contrasting Responses across Identical and Diverse Prompts.*** [Paper] [Code] (* denotes equal contribution)
   - ◇ Generalizes DPO by constructing a **contrast matrix over identical and semantically related prompts** with prompt-similarity-based weighting, enabling preference learning from both paired and unpaired data.
   - ◇ Outperforms DPO, IPO, and KTO on Anthropic-HH, OpenAI Summarization, and **AlpacaEval 2.0**, demonstrating stronger alignment across dialogue and summarization tasks.
2. **Yueqin Yin***, Zhendong Wang*, Yujia Xie, Weizhu Chen, Mingyuan Zhou. ***Self-Augmented Preference Optimization: Off-Policy Paradigms for Language Model Alignment.*** [Paper] [Code] (* denotes equal contribution)
   - ◇ Proposes an **off-policy self-play** alignment framework that eliminates the need for paired preference data or external reward models, combining an EMA model with a replay buffer for dynamic, curriculum-style negative sample generation.
   - ◇ Matches or exceeds DPO, ORPO, and SPIN across **Open LLM Leaderboard, IFEval, MT-Bench, and AlpacaEval 2.0**, demonstrating scalable alignment without external supervision.
3. Yi Gu, Zhendong Wang, **Yueqin Yin**, Yujia Xie, Mingyuan Zhou. ***Diffusion-RPO: Aligning Diffusion Models through Relative Preference Optimization.*** [Paper]
   - ◇ Extends RPO to text-to-image diffusion models with a **step-wise contrastive weighting** mechanism, contrasting both identical and semantically related prompt–image pairs for fine-grained visual preference alignment.
   - ◇ Introduces a new **Style Alignment** evaluation task; achieves superior performance over Diffusion-DPO and SFT on Stable Diffusion 1.5 and XL across human preference and style benchmarks.

Internship & Project Experience

Generative Adversarial Network (GAN) Models. — Shanghai AI Laboratory
Mar. 2021 – Dec. 2021 / Advisors: Qianyi Wu, Wayne Wu — *Research Intern, Shanghai, China*

- ◇ **Talking Face Generation.** Developed a neural system that generates talking faces driven by audio signals. Based on the First Order Motion Model (FOMM), replaced the image-driven branch with an *MFCC-based motion encoder*, allowing dense optical flow fields to warp source features for synchronized lip and facial movement synthesis.
- ◇ **GAN-based Image Generation and Editing.** Introduced a **cross-space interaction mechanism** between the P-space and Z-space of StyleGAN for disentangled and controllable facial attribute editing, achieving SOTA controllability and identity preservation. Published at **CVPR 2022**.

Diffusion Models. — Alibaba DAMO Academy
Mar. 2022 – Mar. 2023 / Advisors: Lianghua Huang, Yu Liu — *Research Intern, Beijing, China*

- ◇ **Continuous Diffusion Models.** Introduced the **generative artifact restoration task** with two new datasets, and developed **DiffGAR**, a conditional diffusion framework that restores clean images from artifact-contaminated GAN/diffusion outputs. Published at **CSAI 2022**.
- ◇ **Discrete Diffusion Models.** Proposed an improved prior modeling approach for VQ-Diffusion by designing a **two-layer VQGAN structure** that expands the single-layer Markov chain into a hierarchical two-layer process. Incorporated an auxiliary supervision loss on 32×32 discrete tokens to enhance prior learning on 16×16 tokens with minimal computational overhead.
- ◇ **Transformer-Based Two-Stage Vector-Quantized Model.** Designed a **hybrid image generation pipeline** combining autoencoding-based training with autoregressive inference, achieving higher fidelity and faster inference than prior two-stage methods. Published at **ICCV 2023**.

Large Language Model Alignment. — The University of Texas at Austin
Dec. 2023 – May. 2024 / Advisor: Mingyuan Zhou — *Research Assistant, Austin, TX*

- **Relative Preference Optimization (RPO).** Generalized DPO with a contrastive framework over identical and semantically related prompts, enabling preference learning from both paired and unpaired data. Outperforms DPO, IPO, and KTO on Anthropic-HH, OpenAI Summarization, and AlpacaEval 2.0.
- **Self-Augmented Preference Optimization (SAPO).** Designed an off-policy self-play alignment framework using an EMA model and replay buffer for dynamic negative sample generation, eliminating the need for paired data or external reward models. Achieves SOTA on AlpacaEval 2.0, IFEval, and MT-Bench, surpassing DPO, ORPO, and SPIN.

Large Language Model Alignment and Reasoning. Microsoft Research
June. 2024 – May. 2025 / Mentors: Yujia Xie, Ziyi Yang *Research Intern, Redmond, WA*

- **Segment-Level PPO.** Proposed a segment-level reward model using entropy-based text segmentation with location-aware normalization and reward interpolation for dense, stable PPO training. Achieves competitive results on AlpacaEval 2.0, Arena-Hard, and MT-Bench. Accepted by **TMLR 2025**.
- **KODCODE.** Created a 447K-problem synthetic coding dataset with self-verified solutions and unit tests across 12 diverse sources (<2.5% error rate). Fine-tuned models achieve SOTA on HumanEval(+), MBPP(+), BigCodeBench, and LiveCodeBench. **ACL 2025 Findings**; 🏆 **Best Paper Award at DataWorld @ ICML 2025**.

Hallucination Detection and Verification. Zoom Video Communications, GenAI Research Group
June 2025 – Aug. 2025 / Mentor: Song Wang *Research Intern, Bellevue, WA*

- **ContextCheck.** Developed a context-conditioned faithfulness verifier that resolves pronouns and references by conditioning on preceding discourse, eliminating separate decontextualization steps. Achieves SOTA on 3 context-dependent benchmarks (**10+ point Macro F1 gain**) and 14 standard single-sentence datasets. Accepted by **ACL 2026 Findings**.

SQL Code Completion. Snowflake AI Research
Jun. 2026 – Present / Mentors: Yite Wang, Zhewei Yao *Research Intern, Bellevue, WA*

- **Fill-in-the-Middle SQL Coding Model.** Working on improving fill-in-the-middle (FIM) models for SQL code completion, focusing on constructing higher-quality supervised fine-tuning data from production usage logs and exploring reinforcement learning reward design to enhance suggestion accuracy and acceptance rates.

Academic Service

Reviewer for **CVPR 2024**, **ICCV 2025**, **ICLR 2025**, **ICLR 2026**.